

# Model for “Te”

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## General Condition

- Basis type: **1g**
- SAMB selection:
  - Type: **[Q, G]**
  - Rank: **[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]**
  - Irrep.: **[A<sub>1</sub>, A<sub>2</sub>, E]**
  - Spin (s): **[0, 1]**
- Atomic selection:
  - Type: **[Q, G, M, T]**
  - Rank: **[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]**
  - Irrep.: **[A<sub>1</sub>, A<sub>2</sub>, E]**
  - Spin (s): **[0, 1]**
- Site-cluster selection:
  - Rank: **[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]**
  - Irrep.: **[A<sub>1</sub>, A<sub>2</sub>, E]**
- Bond-cluster selection:
  - Type: **[Q, G, M, T]**
  - Rank: **[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]**
  - Irrep.: **[A<sub>1</sub>, A<sub>2</sub>, E]**
- Max. neighbor: **10**
- Search cell range: **(-2, 3), (-2, 3), (-2, 3)**
- Toroidal priority: **false**

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## Group and Unit Cell

- Group: SG No. 152  $D_3^4$   $P3_121$  [ trigonal ]
- Associated point group: PG No. 152  $D_3$  32 (321 setting) [ trigonal ]
- Unit cell:
  - $a = 4.45800$ ,  $b = 4.45800$ ,  $c = 5.92500$ ,  $\alpha = 90.0$ ,  $\beta = 90.0$ ,  $\gamma = 120.0$
- Lattice vectors (conventional cell):
  - $\mathbf{a}_1 = [ 4.45800, 0.00000, 0.00000 ]$
  - $\mathbf{a}_2 = [ -2.22900, 3.86074, 0.00000 ]$
  - $\mathbf{a}_3 = [ 0.00000, 0.00000, 5.92500 ]$

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## Symmetry Operation

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Table 1: Symmetry operation

| # | SO                          | # | SO                            | # | SO                            | # | SO              | # | SO                          |
|---|-----------------------------|---|-------------------------------|---|-------------------------------|---|-----------------|---|-----------------------------|
| 1 | $\{1 0\}$                   | 2 | $\{3_{001}^+ 00\frac{1}{3}\}$ | 3 | $\{3_{001}^- 00\frac{2}{3}\}$ | 4 | $\{2_{110} 0\}$ | 5 | $\{2_{100} 00\frac{2}{3}\}$ |
| 6 | $\{2_{010} 00\frac{1}{3}\}$ |   |                               |   |                               |   |                 |   |                             |

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## Harmonics

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Table 2: Harmonics

| # | symbol                 | irrep. | rank | X      | multiplicity | component | symmetry                               |
|---|------------------------|--------|------|--------|--------------|-----------|----------------------------------------|
| 1 | $\mathbb{G}_0(A_1)$    | $A_1$  | 0    | $G, M$ | -            | -         | 1                                      |
| 2 | $\mathbb{Q}_0(A_1)$    | $A_1$  | 0    | $Q, T$ | -            | -         | 1                                      |
| 3 | $\mathbb{G}_2(A_1)$    | $A_1$  | 2    | $G, M$ | -            | -         | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| 4 | $\mathbb{Q}_2(A_1)$    | $A_1$  | 2    | $Q, T$ | -            | -         | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| 5 | $\mathbb{Q}_3(A_1)$    | $A_1$  | 3    | $Q, T$ | -            | -         | $\frac{\sqrt{10}x(x^2-3y^2)}{4}$       |
| 6 | $\mathbb{G}_1(A_2)$    | $A_2$  | 1    | $G, M$ | -            | -         | $z$                                    |
| 7 | $\mathbb{Q}_1(A_2)$    | $A_2$  | 1    | $Q, T$ | -            | -         | $z$                                    |
| 8 | $\mathbb{Q}_3(A_2, 2)$ | $A_2$  | 3    | $Q, T$ | 2            | -         | $\frac{\sqrt{10}y(3x^2-y^2)}{4}$       |
| 9 | $\mathbb{G}_{1,1}(E)$  | $E$    | 1    | $G, M$ | -            | 1         | $x$                                    |

*continued ...*

Table 2

| #  | symbol                   | irrep. | rank | X      | multiplicity | component | symmetry                             |
|----|--------------------------|--------|------|--------|--------------|-----------|--------------------------------------|
| 10 | $\mathbb{G}_{1,2}(E)$    |        |      |        |              | 2         | $y$                                  |
| 11 | $\mathbb{Q}_{1,1}(E)$    | $E$    | 1    | $Q, T$ | -            | 1         | $x$                                  |
| 12 | $\mathbb{Q}_{1,2}(E)$    |        |      |        |              | 2         | $y$                                  |
| 13 | $\mathbb{G}_{2,1}(E, 1)$ | $E$    | 2    | $G, M$ | 1            | 1         | $\sqrt{3}yz$                         |
| 14 | $\mathbb{G}_{2,2}(E, 1)$ |        |      |        |              | 2         | $-\sqrt{3}xz$                        |
| 15 | $\mathbb{G}_{2,1}(E, 2)$ | $E$    | 2    | $G, M$ | 2            | 1         | $\frac{\sqrt{3}(x-y)(x+y)}{2}$       |
| 16 | $\mathbb{G}_{2,2}(E, 2)$ |        |      |        |              | 2         | $-\sqrt{3}xy$                        |
| 17 | $\mathbb{Q}_{2,1}(E, 1)$ | $E$    | 2    | $Q, T$ | 1            | 1         | $\sqrt{3}yz$                         |
| 18 | $\mathbb{Q}_{2,2}(E, 1)$ |        |      |        |              | 2         | $-\sqrt{3}xz$                        |
| 19 | $\mathbb{Q}_{2,1}(E, 2)$ | $E$    | 2    | $Q, T$ | 2            | 1         | $\frac{\sqrt{3}(x-y)(x+y)}{2}$       |
| 20 | $\mathbb{Q}_{2,2}(E, 2)$ |        |      |        |              | 2         | $-\sqrt{3}xy$                        |
| 21 | $\mathbb{Q}_{3,1}(E, 1)$ | $E$    | 3    | $Q, T$ | 1            | 1         | $-\frac{\sqrt{6}x(x^2+y^2-4z^2)}{4}$ |
| 22 | $\mathbb{Q}_{3,2}(E, 1)$ |        |      |        |              | 2         | $-\frac{\sqrt{6}y(x^2+y^2-4z^2)}{4}$ |
| 23 | $\mathbb{Q}_{3,1}(E, 2)$ | $E$    | 3    | $Q, T$ | 2            | 1         | $\sqrt{15}xyz$                       |
| 24 | $\mathbb{Q}_{3,2}(E, 2)$ |        |      |        |              | 2         | $\frac{\sqrt{15}z(x-y)(x+y)}{2}$     |

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 Basis in full matrix
 

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Table 3: dimension = 9

| # | orbital@atom(SL)    | # | orbital@atom(SL)    | # | orbital@atom(SL)    | # | orbital@atom(SL)    | # | orbital@atom(SL)    |
|---|---------------------|---|---------------------|---|---------------------|---|---------------------|---|---------------------|
| 0 | $ p_x\rangle @A(1)$ | 1 | $ p_y\rangle @A(1)$ | 2 | $ p_z\rangle @A(1)$ | 3 | $ p_x\rangle @A(2)$ | 4 | $ p_y\rangle @A(2)$ |
| 5 | $ p_z\rangle @A(2)$ | 6 | $ p_x\rangle @A(3)$ | 7 | $ p_y\rangle @A(3)$ | 8 | $ p_z\rangle @A(3)$ |   |                     |

Table 4: Atomic basis (orbital part only)

| orbital       | definition |
|---------------|------------|
| $ p_x\rangle$ | $x$        |
| $ p_y\rangle$ | $y$        |
| $ p_z\rangle$ | $z$        |

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**SAMB: 45 (all 45)**


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- **A** : 'A' site-cluster

- \* bra:  $\langle p_x|, \langle p_y|, \langle p_z|$

- \* ket:  $|p_x\rangle, |p_y\rangle, |p_z\rangle$

- \* wyckoff: **3a**

$$\boxed{\text{z1}} \quad Q_0^{(c)}(A_1) = Q_0^{(a)}(A_1)Q_0^{(s)}(A_1)$$

$$\boxed{\text{z2}} \quad Q_2^{(c)}(A_1) = Q_2^{(a)}(A_1)Q_0^{(s)}(A_1)$$

$$\boxed{\text{z3}} \quad Q_3^{(c)}(A_1) = \frac{\sqrt{2}Q_{2,1}^{(a)}(E, 2)Q_{1,1}^{(s)}(E)}{2} + \frac{\sqrt{2}Q_{2,2}^{(a)}(E, 2)Q_{1,2}^{(s)}(E)}{2}$$

$$\boxed{\text{z4}} \quad \mathbb{G}_2^{(c)}(A_1) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E, 1)\mathbb{Q}_{1,1}^{(s)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E, 1)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$$

$$\boxed{\text{z11}} \quad \mathbb{Q}_1^{(c)}(A_2) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E, 1)\mathbb{Q}_{1,2}^{(s)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E, 1)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$$

$$\boxed{\text{z12}} \quad \mathbb{Q}_3^{(c)}(A_2, 2) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E, 2)\mathbb{Q}_{1,2}^{(s)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E, 2)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$$

$$\boxed{\text{z16}} \quad \mathbb{Q}_{1,1}^{(c)}(E, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$$

$$\boxed{\text{z17}} \quad \mathbb{Q}_{1,2}^{(c)}(E, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$$

$$\boxed{\text{z18}} \quad \mathbb{Q}_{1,1}^{(c)}(E, b) = \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(E, 2)\mathbb{Q}_{1,1}^{(s)}(E)}{14} - \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(E, 2)\mathbb{Q}_{1,2}^{(s)}(E)}{14} - \frac{\sqrt{14}\mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(E)}{14}$$

$$\boxed{\text{z19}} \quad \mathbb{Q}_{1,2}^{(c)}(E, b) = -\frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(E, 2)\mathbb{Q}_{1,2}^{(s)}(E)}{14} - \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(E, 2)\mathbb{Q}_{1,1}^{(s)}(E)}{14} - \frac{\sqrt{14}\mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(E)}{14}$$

$$\boxed{\text{z20}} \quad \mathbb{Q}_{2,1}^{(c)}(E, 1) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E, 1)\mathbb{Q}_0^{(s)}(A_1)}{2}$$

$$\boxed{\text{z21}} \quad \mathbb{Q}_{2,2}^{(c)}(E, 1) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E, 1)\mathbb{Q}_0^{(s)}(A_1)}{2}$$

$$\boxed{\text{z22}} \quad \mathbb{Q}_{2,1}^{(c)}(E, 2) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E, 2)\mathbb{Q}_0^{(s)}(A_1)}{2}$$

$$\boxed{\text{z23}} \quad \mathbb{Q}_{2,2}^{(c)}(E, 2) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E, 2)\mathbb{Q}_0^{(s)}(A_1)}{2}$$

$$\boxed{\text{z24}} \quad \mathbb{Q}_{3,1}^{(c)}(E, 1) = \frac{\sqrt{7}\mathbb{Q}_{2,1}^{(a)}(E, 2)\mathbb{Q}_{1,1}^{(s)}(E)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(a)}(E, 2)\mathbb{Q}_{1,2}^{(s)}(E)}{14} + \frac{\sqrt{21}\mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_{1,1}^{(s)}(E)}{7}$$

$$\boxed{\text{z25}} \quad \mathbb{Q}_{3,2}^{(c)}(E, 1) = -\frac{\sqrt{7}\mathbb{Q}_{2,1}^{(a)}(E, 2)\mathbb{Q}_{1,2}^{(s)}(E)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(a)}(E, 2)\mathbb{Q}_{1,1}^{(s)}(E)}{14} + \frac{\sqrt{21}\mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_{1,2}^{(s)}(E)}{7}$$

$$\boxed{\text{z26}} \quad \mathbb{Q}_{3,1}^{(c)}(E, 2) = \frac{\mathbb{Q}_{2,1}^{(a)}(E, 1)\mathbb{Q}_{1,1}^{(s)}(E)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E, 1)\mathbb{Q}_{1,2}^{(s)}(E)}{2}$$

$$\boxed{\text{z27}} \quad \mathbb{Q}_{3,2}^{(c)}(E, 2) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E, 1)\mathbb{Q}_{1,2}^{(s)}(E)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E, 1)\mathbb{Q}_{1,1}^{(s)}(E)}{2}$$

• **A;A\_001\_1** : 'A'-'A' bond-cluster

\* bra:  $\langle p_x |, \langle p_y |, \langle p_z |$

\* ket:  $|p_x\rangle, |p_y\rangle, |p_z\rangle$

\* wyckoff: **3a@3b**

$$\boxed{\text{z5}} \quad \mathbb{Q}_0^{(c)}(A_1) = \mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_0^{(b)}(A_1)$$

$$\boxed{\text{z6}} \quad \mathbb{Q}_2^{(c)}(A_1) = \mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_0^{(b)}(A_1)$$

$$\boxed{\text{z7}} \quad \mathbb{Q}_3^{(c)}(A_1) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E, 2)\mathbb{Q}_{1,1}^{(b)}(E)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E, 2)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$$\boxed{\text{z8}} \quad \mathbb{G}_0^{(c)}(A_1) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{3} + \frac{\sqrt{3}\mathbb{M}_1^{(a)}(A_2)\mathbb{T}_1^{(b)}(A_2)}{3}$$

$$\boxed{\text{z9}} \quad \mathbb{G}_2^{(c)}(A_1, a) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E, 1)\mathbb{Q}_{1,1}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E, 1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$$\boxed{\text{z10}} \quad \mathbb{G}_2^{(c)}(A_1, b) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{6} + \frac{\sqrt{6}\mathbb{M}_1^{(a)}(A_2)\mathbb{T}_1^{(b)}(A_2)}{3}$$

$$\boxed{\text{z13}} \quad \mathbb{Q}_1^{(c)}(A_2, a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E, 1)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E, 1)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$$\boxed{\text{z14}} \quad \mathbb{Q}_1^{(c)}(A_2, b) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

$$\boxed{\text{z15}} \quad \mathbb{Q}_3^{(c)}(A_2, 2) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E, 2)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E, 2)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$$\boxed{\text{z28}} \quad \mathbb{Q}_{1,1}^{(c)}(E, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{2}$$

$$\boxed{\text{z29}} \quad \mathbb{Q}_{1,2}^{(c)}(E, a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{2}$$

$$\begin{aligned}
\boxed{\text{z30}} \quad \mathbb{Q}_{1,1}^{(c)}(E, b) &= \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(E, 2)\mathbb{Q}_{1,1}^{(b)}(E)}{14} - \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(E, 2)\mathbb{Q}_{1,2}^{(b)}(E)}{14} - \frac{\sqrt{14}\mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{14} \\
\boxed{\text{z31}} \quad \mathbb{Q}_{1,2}^{(c)}(E, b) &= -\frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(E, 2)\mathbb{Q}_{1,2}^{(b)}(E)}{14} - \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(E, 2)\mathbb{Q}_{1,1}^{(b)}(E)}{14} - \frac{\sqrt{14}\mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{14} \\
\boxed{\text{z32}} \quad \mathbb{Q}_{1,1}^{(c)}(E, c) &= \frac{\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_1^{(b)}(A_2)}{2} - \frac{\mathbb{M}_1^{(a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2} \\
\boxed{\text{z33}} \quad \mathbb{Q}_{1,2}^{(c)}(E, c) &= -\frac{\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_1^{(b)}(A_2)}{2} + \frac{\mathbb{M}_1^{(a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2} \\
\boxed{\text{z34}} \quad \mathbb{Q}_{2,1}^{(c)}(E, 1) &= \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E, 1)\mathbb{Q}_0^{(b)}(A_1)}{2} \\
\boxed{\text{z35}} \quad \mathbb{Q}_{2,2}^{(c)}(E, 1) &= \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E, 1)\mathbb{Q}_0^{(b)}(A_1)}{2} \\
\boxed{\text{z36}} \quad \mathbb{Q}_{2,1}^{(c)}(E, 2) &= \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E, 2)\mathbb{Q}_0^{(b)}(A_1)}{2} \\
\boxed{\text{z37}} \quad \mathbb{Q}_{2,2}^{(c)}(E, 2) &= \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E, 2)\mathbb{Q}_0^{(b)}(A_1)}{2} \\
\boxed{\text{z38}} \quad \mathbb{Q}_{3,1}^{(c)}(E, 1) &= \frac{\sqrt{7}\mathbb{Q}_{2,1}^{(a)}(E, 2)\mathbb{Q}_{1,1}^{(b)}(E)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(a)}(E, 2)\mathbb{Q}_{1,2}^{(b)}(E)}{14} + \frac{\sqrt{21}\mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_{1,1}^{(b)}(E)}{7} \\
\boxed{\text{z39}} \quad \mathbb{Q}_{3,2}^{(c)}(E, 1) &= -\frac{\sqrt{7}\mathbb{Q}_{2,1}^{(a)}(E, 2)\mathbb{Q}_{1,2}^{(b)}(E)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(a)}(E, 2)\mathbb{Q}_{1,1}^{(b)}(E)}{14} + \frac{\sqrt{21}\mathbb{Q}_2^{(a)}(A_1)\mathbb{Q}_{1,2}^{(b)}(E)}{7} \\
\boxed{\text{z40}} \quad \mathbb{Q}_{3,1}^{(c)}(E, 2) &= \frac{\mathbb{Q}_{2,1}^{(a)}(E, 1)\mathbb{Q}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E, 1)\mathbb{Q}_{1,2}^{(b)}(E)}{2} \\
\boxed{\text{z41}} \quad \mathbb{Q}_{3,2}^{(c)}(E, 2) &= -\frac{\mathbb{Q}_{2,1}^{(a)}(E, 1)\mathbb{Q}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E, 1)\mathbb{Q}_{1,1}^{(b)}(E)}{2} \\
\boxed{\text{z42}} \quad \mathbb{G}_{2,1}^{(c)}(E, 1) &= \frac{\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_1^{(b)}(A_2)}{2} + \frac{\mathbb{M}_1^{(a)}(A_2)\mathbb{T}_{1,2}^{(b)}(E)}{2} \\
\boxed{\text{z43}} \quad \mathbb{G}_{2,2}^{(c)}(E, 1) &= -\frac{\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_1^{(b)}(A_2)}{2} - \frac{\mathbb{M}_1^{(a)}(A_2)\mathbb{T}_{1,1}^{(b)}(E)}{2}
\end{aligned}$$

$$\boxed{\text{z44}} \quad \mathbb{G}_{2,1}^{(c)}(E, 2) = \frac{\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2}$$

$$\boxed{\text{z45}} \quad \mathbb{G}_{2,2}^{(c)}(E, 2) = -\frac{\mathbb{M}_{1,1}^{(a)}(E)\mathbb{T}_{1,2}^{(b)}(E)}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(E)\mathbb{T}_{1,1}^{(b)}(E)}{2}$$

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### Atomic SAMB

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- bra:  $\langle p_x |, \langle p_y |, \langle p_z |$
- ket:  $|p_x\rangle, |p_y\rangle, |p_z\rangle$

$$\boxed{\text{x1}} \quad \mathbb{Q}_0^{(a)}(A_1) = \begin{bmatrix} \frac{\sqrt{3}}{3} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{3} & 0 \\ 0 & 0 & \frac{\sqrt{3}}{3} \end{bmatrix}$$

$$\boxed{\text{x2}} \quad \mathbb{Q}_2^{(a)}(A_1) = \begin{bmatrix} -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{3} \end{bmatrix}$$

$$\boxed{\text{x3}} \quad \mathbb{Q}_{2,1}^{(a)}(E, 1) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$$

$$\boxed{\text{x4}} \quad \mathbb{Q}_{2,2}^{(a)}(E, 1) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 0 & 0 \\ -\frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x5}} \quad \mathbb{Q}_{2,1}^{(a)}(E, 2) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



$$\boxed{\text{x6}} \quad \mathbb{Q}_{2,2}^{(a)}(E, 2) = \begin{bmatrix} 0 & -\frac{\sqrt{2}}{2} & 0 \\ -\frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x7}} \quad \mathbb{M}_1^{(a)}(A_2) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} & 0 \\ \frac{\sqrt{2}i}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x8}} \quad \mathbb{M}_{1,1}^{(a)}(E) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{2} \\ 0 & \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$$

$$\boxed{\text{x9}} \quad \mathbb{M}_{1,2}^{(a)}(E) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{2} & 0 & 0 \end{bmatrix}$$

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### Cluster SAMB

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- Site cluster

\*\* Wyckoff: 3a

$$\boxed{\text{y1}} \quad \mathbb{Q}_0^{(s)}(A_1) = \left[ \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3} \right]$$

$$\boxed{\text{y2}} \quad \mathbb{Q}_{1,1}^{(s)}(E) = \left[ \frac{\sqrt{6}}{3}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6} \right]$$

$$\boxed{\text{y3}} \quad \mathbb{Q}_{1,2}^{(s)}(E) = \left[ 0, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2} \right]$$

- Bond cluster

\*\* Wyckoff: 3a@3b

$$\boxed{\text{y4}} \quad \mathbb{Q}_0^{(b)}(A_1) = \left[ \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3} \right]$$

$$\boxed{\text{y5}} \quad \mathbb{T}_1^{(b)}(A_2) = \left[ \frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{3} \right]$$

$$\boxed{\text{y6}} \quad \mathbb{Q}_{1,1}^{(b)}(E) = \left[ \frac{\sqrt{6}}{3}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6} \right]$$

$$\boxed{\text{y7}} \quad \mathbb{Q}_{1,2}^{(b)}(E) = \left[ 0, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2} \right]$$

$$\boxed{\text{y8}} \quad \mathbb{T}_{1,1}^{(b)}(E) = \left[ 0, -\frac{\sqrt{2}i}{2}, \frac{\sqrt{2}i}{2} \right]$$

$$\boxed{\text{y9}} \quad \mathbb{T}_{1,2}^{(b)}(E) = \left[ \frac{\sqrt{6}i}{3}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6} \right]$$

— Site and Bond —

Table 5: Orbital of each site

| # | site     | orbital                                 |
|---|----------|-----------------------------------------|
| 1 | <b>A</b> | $ p_x\rangle,  p_y\rangle,  p_z\rangle$ |

Table 6: Neighbor and bra-ket of each bond

| # | head | tail | neighbor | head (bra) | tail (ket) |
|---|------|------|----------|------------|------------|
| 1 | A    | A    | [1]      | [p]        | [p]        |

#### — Site in Unit Cell —

Sites in (conventional) cell (no plus set), SL = sublattice

Table 7: 'A' (#1) site cluster (3a), .2.

| SL | position ( $s$ )             | mapping |
|----|------------------------------|---------|
| 1  | [ 0.27400, 0.00000, 0.33333] | [1,5]   |
| 2  | [ 0.00000, 0.27400, 0.66667] | [2,4]   |
| 3  | [ 0.72600, 0.72600, 0.00000] | [3,6]   |

#### — Bond in Unit Cell —

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 5th neighbor at most)

Table 8: 1-th 'A'-'A' [1] (#1) bond cluster (3a@3b), ND,  $|v|=2.89426$  (cartesian)

| SL | vector ( $v$ )                | center ( $c$ )               | mapping | head   | tail   | $R$ (primitive) |
|----|-------------------------------|------------------------------|---------|--------|--------|-----------------|
| 1  | [-0.27400, -0.54800, 0.33333] | [ 0.86300, 0.00000, 0.83333] | [1, -5] | (3, 1) | (2, 1) | [1, 1, -1]      |
| 2  | [ 0.54800, 0.27400, 0.33333]  | [ 0.00000, 0.86300, 0.16667] | [2, -4] | (1, 1) | (3, 1) | [-1, -1, 0]     |
| 3  | [-0.27400, 0.27400, 0.33333]  | [ 0.13700, 0.13700, 0.50000] | [3, -6] | (2, 1) | (1, 1) | [0, 0, 0]       |